

Jonathan Jacob Koshy

Portfolio | johnkoper12@gmail.com | LinkedIn | GitHub: JJKSweaty

Education

University of Waterloo
BASc. Electrical Engineering

Waterloo, ON
Expected 2029

Experience

Embedded Software Intern – Belimo - Montreal, QC May 2026 – Aug 2026

- Developed bare metal **Embedded C** on **8 bit PIC18 MCU**, reducing HVAC controller memory use by **20%**
- Validated **CAN, UART, and I2C** signals with lab tools to isolate timing faults and controller state bugs
- Optimized interrupt timing and debounce windows, improving keypad navigation latency by **30%**
- Implemented **ADC digital filtering** for gas sensing firmware, improving measurement accuracy by **10%**
- Improved **BACnet** communication by debugging UART parsing, frame handling, and state recovery faults
- Tested firmware against UL verification guidelines across sensor behavior, edge cases, and PCB level validation

Embedded Software Intern – AeroCardia - Montreal, QC Sep 2025 – Dec 2025

- Developed **C++ sensor** drivers for IMU, PPG, and temperature sensors, enabling **50 Hz** biosignal capture
- Built **RTOS** pipeline using queues to buffer biosensor data before **BLE** transmission, resulting in **<1%** packet loss
- Optimized sensor sampling and buffering paths, reducing dropped samples by **25%** during continuous monitoring
- Implemented secure **BLE OTA** updates with image verification and rollback, cutting update time to **under 1 min**
- Designed an O2 sensor PCB in **Altium** with onboard EEPROM for persistent sensor calibration data

Embedded Software Member – UWARG – Waterloo, ON Dec 2024 – Sep 2025

- Implemented roll and yaw motor mixing firmware, mapping pilot inputs to motor outputs in fixed-wing modes
- Tuned **PID** attitude loops, reducing manual flight oscillation by **20%** during stabilization tests
- Improved control loop timing and actuator output scaling, reducing motor command jitter by **30%**

Embedded Software Intern – Electrium Mobility – Waterloo, ON May 2025 – Aug 2025

- Built **ESP32** firmware for an electric skateboard dashboard, streaming **BLE** telemetry using FreeRTOS
- Integrated **VESC** motor controller data over **UART**, streaming RPM, current, and battery metrics
- Added packet parsing and checksum validation for skateboard diagnostics, reducing malformed frames by **25%**

Projects

Custom VR/AR Headset C++, STM32F411, ICM20948, I2C, USB HID, FastIMU, KiCad 

- Developed STM32F4 firmware in C interfacing 9DoF IMU over I2C, handling sensor reads, scaling, and timing
- Implemented sensor fusion using FastIMU to compute stable orientation (quaternion/Euler) for real-time tracking
- Built USB HID interface to stream head pose data to PC, enabling integration with SteamVR tracking pipeline
- Designed custom KiCad PCB with USB-C power/data, 3.3V regulation, IMU routing, and bring-up test points

Edge AI Dementia Companion (HackCanada Winner) Python, YOLOv8, OpenCV, TypeScript, RPi 5 

- Built an edge AI memory aid with RPi 5 & Hailo-8L using YOLO to track missing objects and last seen locations
- Added medication checks, object memory, and WebSocket snapshots for phone based caregiver assistance

ESP32 Wi-Fi Media Controller C++, Python, ESP32, LVGL, Wi-Fi, WebSockets  

- Built a custom handheld media dashboard with **LVGL** for PC metrics, media state, and Discord data
- Designed a TCP JSON protocol with queued updates, RGB565 artwork rendering, and touchscreen media controls

User Space TCP/IP Stack C, Linux TAP, Ethernet, ARP, IPv4, TCP 

- Built a userspace network stack in C over Linux TAP, processing raw L2 frames across ARP, IPv4, and TCP
- Implemented TCB state logic for connection setup/teardown, retransmission, timeouts, and recovery

Skills

- **Programming:** C, C++, Python, RISC-V, MATLAB, Git
- **Protocols/Platforms:** SPI, I2C, UART, BLE, CAN, Ethernet, PWM, RTOS, ARM Cortex, Raspberry Pi, Linux
- **Lab Tools:** Oscilloscopes, Logic Analyzers, Multimeters, Soldering Irons, Bench Power Supplies